

Chapter 17

Cylindrical and Spherical Coordinates

Triple integrals with symmetry, radial geometry, and modern applications

Chapter overview

Rectangular coordinates are natural for boxes, planes, and regions bounded by simple inequalities in x , y , and z . Many important three-dimensional regions are not rectangular. Cylinders, cones, spheres, shells, balls, hemispheres, and rotationally symmetric solids are often much easier to describe using coordinates adapted to their geometry.

This chapter develops two coordinate systems for triple integrals:

- **cylindrical coordinates** (r, θ, z) , which use polar coordinates in the xy -plane and keep the vertical coordinate z ;
- **spherical coordinates** (ρ, θ, ϕ) , which describe a point by distance from the origin and two angles.

The goal is not simply to memorize formulas. The goal is to learn how to choose coordinates that make geometry, integration, probability, and computation simpler.

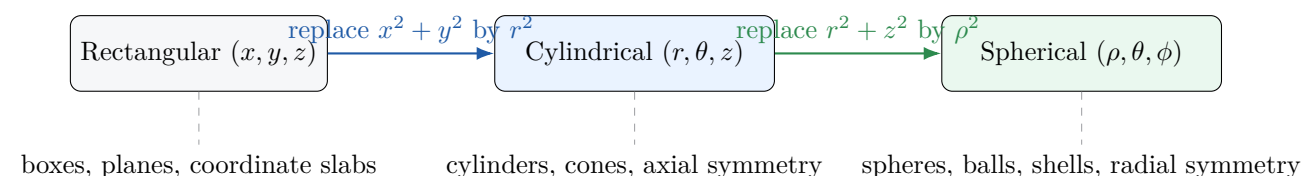


Figure 17.1: Coordinate systems as languages for geometry. The best language is usually the one whose coordinate surfaces match the boundary surfaces.

17.1 Learning goals

After studying this chapter, you should be able to:

1. convert points and equations among rectangular, cylindrical, and spherical coordinates;
2. recognize common coordinate surfaces: cylinders, cones, spheres, and half-planes;
3. use the volume elements $dV = r \, dz \, dr \, d\theta$ and $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$;

4. set up and evaluate triple integrals in cylindrical coordinates;
5. set up and evaluate triple integrals in spherical coordinates;
6. choose an efficient coordinate system by reading the symmetry of the region and integrand;
7. use coordinate transformations in probability, statistics, and simulation.

17.2 Why new coordinates are needed

A coordinate system is a language for describing points. Rectangular coordinates describe a point by three signed distances (x, y, z) . This is perfect for a rectangular box such as

$$0 \leq x \leq 2, \quad 1 \leq y \leq 4, \quad -1 \leq z \leq 3.$$

But the cylinder $x^2 + y^2 \leq 4$ and the sphere $x^2 + y^2 + z^2 \leq 9$ already contain distance expressions. That is the signal that radial coordinates may simplify the problem.

Coordinate choice principle

Use a coordinate system whose coordinate surfaces resemble the boundary surfaces of the region.

- Cylinders suggest cylindrical coordinates.
- Spheres suggest spherical coordinates.
- Cones often work well in either cylindrical or spherical coordinates.
- Planes and boxes often work well in rectangular coordinates.

17.3 Cylindrical coordinates

Cylindrical coordinates

Cylindrical coordinates combine polar coordinates in the xy -plane with the usual vertical coordinate z :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Here $r \geq 0$ and usually $0 \leq \theta \leq 2\pi$. The inverse relations are

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}, \quad z = z.$$

The number r is the distance from (x, y, z) to the z -axis, not the distance to the origin in space.

Cylindrical equation	Geometric meaning	Rectangular form
$r = r_0$	cylinder around the z -axis	$x^2 + y^2 = r_0^2$
$\theta = \theta_0$	vertical half-plane	$y = (\tan \theta_0)x$ with orientation
$z = z_0$	horizontal plane	$z = z_0$
$z = cr$	cone	$z = c\sqrt{x^2 + y^2}$

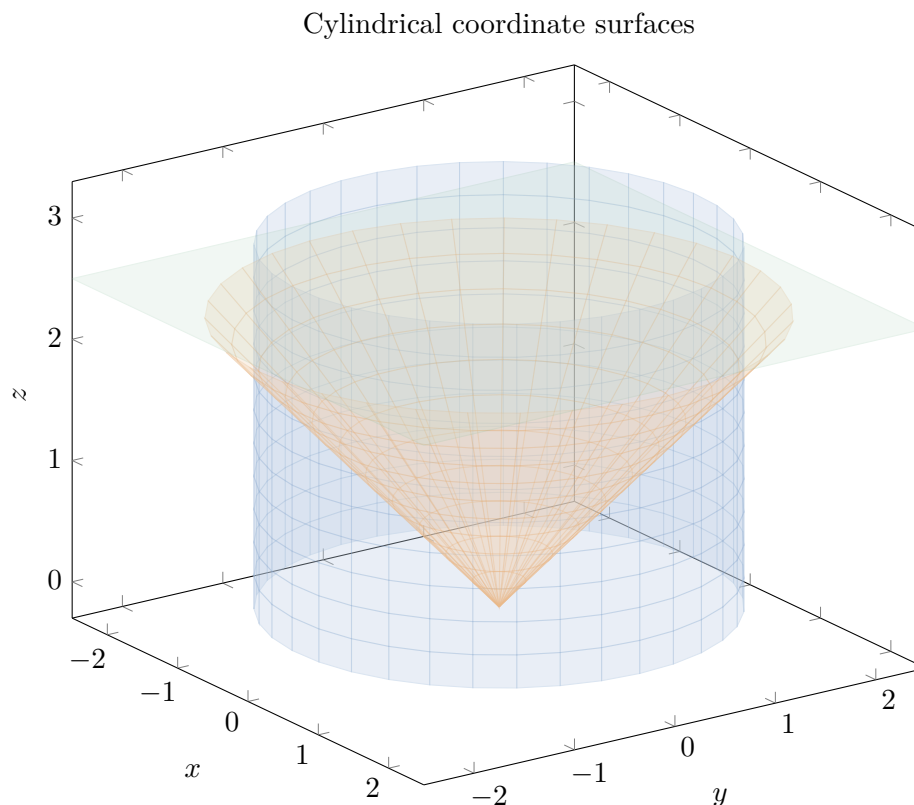


Figure 17.2: Basic cylindrical coordinate surfaces: a cylinder $r = 2$, a cone $z = r$, and a horizontal plane $z = 2.5$.

17.4 The cylindrical volume element

In rectangular coordinates, a small box has volume $dV = dx dy dz$. In cylindrical coordinates, a small coordinate box has dimensions approximately

$$dr, \quad r d\theta, \quad dz.$$

Therefore

$$dV = r dz dr d\theta.$$

The factor r accounts for the fact that an angular change $d\theta$ sweeps out a longer arc when r is larger.

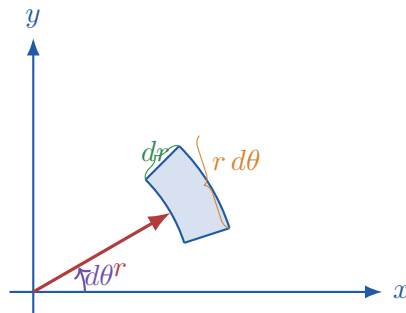


Figure 17.3: The polar base of a cylindrical volume element. Multiplying the base dimensions dr and $r d\theta$ by height dz gives $dV = r dz dr d\theta$.

Cylindrical integration formula

If $x = r \cos \theta$, $y = r \sin \theta$, and $z = z$, then

$$\iiint_E f(x, y, z) dV = \int_{\theta=\alpha}^{\beta} \int_{r=h_1(\theta)}^{h_2(\theta)} \int_{z=g_1(r,\theta)}^{g_2(r,\theta)} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta.$$

The extra factor r is not optional. It is part of the volume element.

17.5 Worked examples in cylindrical coordinates

Example 1: Convert a point

Find rectangular coordinates for $(r, \theta, z) = \left(2, \frac{2\pi}{3}, 3\right)$. Using $x = r \cos \theta$ and $y = r \sin \theta$,

$$x = 2 \cos \left(\frac{2\pi}{3}\right) = -1, \quad y = 2 \sin \left(\frac{2\pi}{3}\right) = \sqrt{3},$$

and $z = 3$. Thus the rectangular coordinates are $(-1, \sqrt{3}, 3)$.

Example 2: A cone below a plane

Find the volume of the solid above the cone $z = r$ and below the plane $z = 1$. The cone and plane intersect when $r = 1$. The region is

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 1, \quad r \leq z \leq 1.$$

Thus

$$V = \int_0^{2\pi} \int_0^1 \int_r^1 r dz dr d\theta = 2\pi \int_0^1 r(1-r) dr = \frac{\pi}{3}.$$

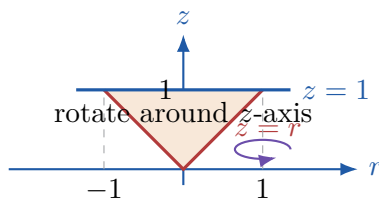


Figure 17.4: A cross-section of the solid above $z = r$ and below $z = 1$. Cylindrical coordinates turn the solid into $0 \leq r \leq 1$, $r \leq z \leq 1$.

Example 3: A cylinder above a paraboloid

Let E be the solid inside $x^2 + y^2 \leq 1$, above the paraboloid $z = 1 - x^2 - y^2$, and below the plane $z = 3$. Since $x^2 + y^2 = r^2$, the bounds are

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 1, \quad 1 - r^2 \leq z \leq 3.$$

Hence

$$V = \int_0^{2\pi} \int_0^1 \int_{1-r^2}^3 r \, dz \, dr \, d\theta = 2\pi \int_0^1 r(2 + r^2) \, dr = \frac{5\pi}{2}.$$

Cylinder, paraboloid, and plane

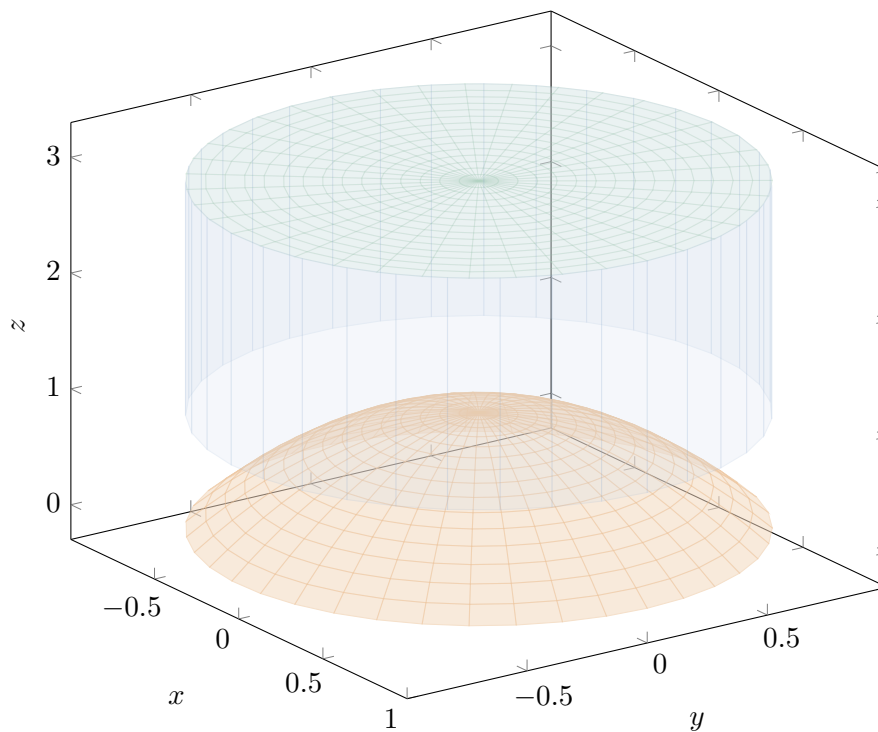


Figure 17.5: A solid inside the cylinder $r = 1$, above $z = 1 - r^2$, and below $z = 3$.

Example 4: Mass with radial density

For the same solid E , suppose $\delta(x, y, z) = \sqrt{x^2 + y^2}$. In cylindrical coordinates, $\delta = r$. Mass is density integrated over volume:

$$M = \int_0^{2\pi} \int_0^1 \int_{1-r^2}^3 r \cdot r \, dz \, dr \, d\theta.$$

The first r is the density and the second r is the Jacobian factor. Therefore

$$M = 2\pi \int_0^1 r^2(2 + r^2) \, dr = 2\pi \left(\frac{2}{3} + \frac{1}{5} \right) = \frac{26\pi}{15}.$$

Example 5: Converting a rectangular integral

Evaluate

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^1 (x^2 + y^2) \, dz \, dy \, dx.$$

The projection onto the xy -plane is the unit disk. The lower surface is the cone $z = \sqrt{x^2 + y^2} = r$, the upper surface is $z = 1$, and the integrand becomes r^2 . Therefore

$$\int_0^{2\pi} \int_0^1 \int_r^1 r^2 \, dz \, dr \, d\theta = 2\pi \int_0^1 r^3(1 - r) \, dr = \frac{\pi}{10}.$$

17.6 Spherical coordinates

Spherical coordinates

Spherical coordinates describe a point by (ρ, θ, ϕ) , where ρ is the distance from the origin, θ is the angle in the xy -plane measured from the positive x -axis, and ϕ is the angle from the positive z -axis. The conversion formulas are

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

Usually $\rho \geq 0$, $0 \leq \theta \leq 2\pi$, and $0 \leq \phi \leq \pi$. The relations with cylindrical coordinates are

$$r = \rho \sin \phi, \quad z = \rho \cos \phi, \quad \theta = \theta.$$

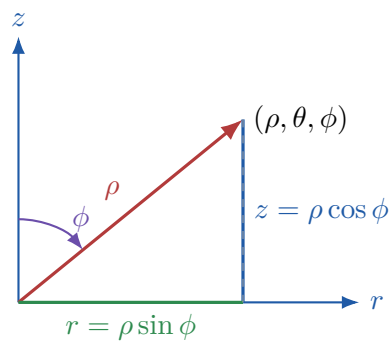


Figure 17.6: In a vertical cross-section through the z -axis, spherical coordinates reduce to right-triangle geometry: $r = \rho \sin \phi$ and $z = \rho \cos \phi$.

Spherical equation	Geometric meaning	Rectangular or cylindrical form
$\rho = \rho_0$	sphere centered at origin	$x^2 + y^2 + z^2 = \rho_0^2$
$\theta = \theta_0$	vertical half-plane	same as cylindrical angle
$\phi = \phi_0$	cone with vertex at origin	$z = r \cot \phi_0$

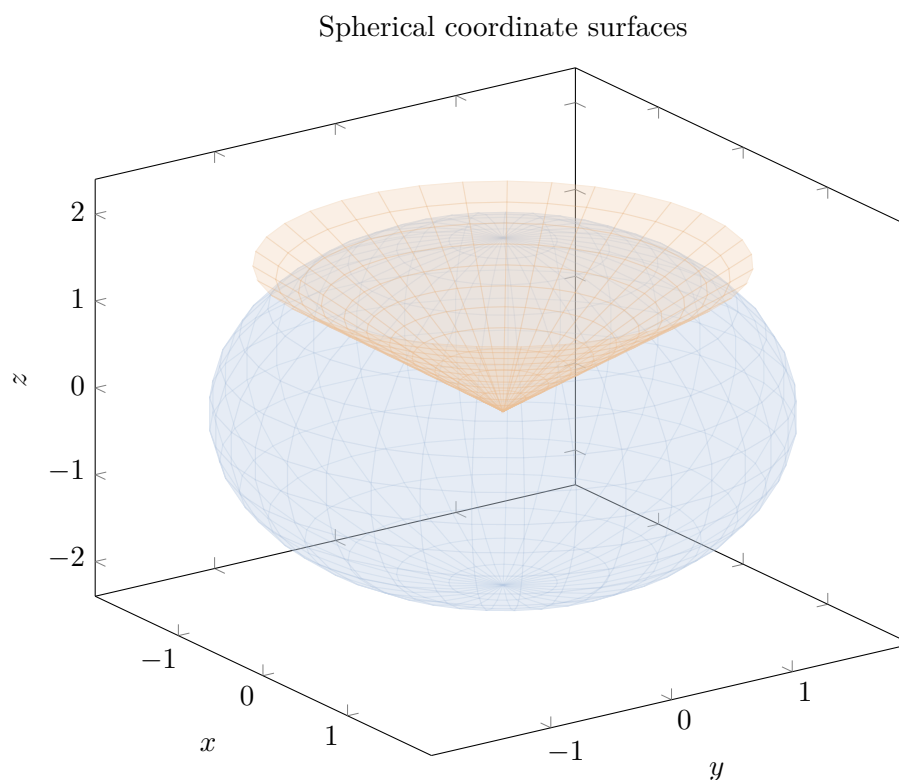


Figure 17.7: Basic spherical coordinate surfaces: the sphere $\rho = 2$ and the cone $\phi = \pi/4$.

17.7 The spherical volume element

A small spherical coordinate box has approximate side lengths

$$d\rho, \quad \rho d\phi, \quad \rho \sin \phi d\theta.$$

Therefore

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta.$$

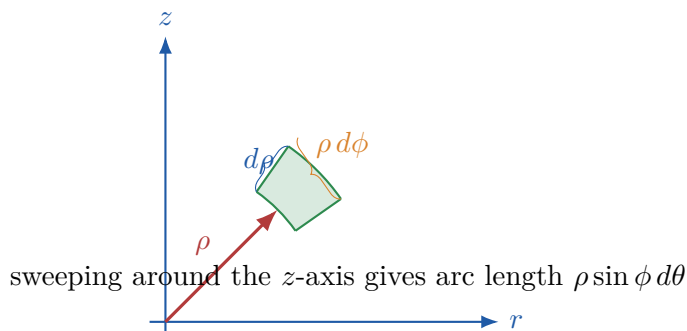


Figure 17.8: The spherical volume factor is a product of three local lengths: $d\rho$, $\rho d\phi$, and $\rho \sin \phi d\theta$.

Spherical integration formula

If

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

then

$$\iiint_E f(x, y, z) dV = \iiint f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\phi d\theta.$$

The factor $\rho^2 \sin \phi$ is the three-dimensional Jacobian factor for spherical coordinates.

17.8 Worked examples in spherical coordinates

Example 6: Volume of the unit ball

For the unit ball $x^2 + y^2 + z^2 \leq 1$, spherical bounds are

$$0 \leq \rho \leq 1, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Thus

$$V = \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^2 \sin \phi d\rho d\phi d\theta = (2\pi)(2) \left(\frac{1}{3} \right) = \frac{4\pi}{3}.$$

Example 7: Inside a sphere and above a cone

Find the volume inside $x^2 + y^2 + z^2 = 8$ and above $z = \sqrt{x^2 + y^2}$. The sphere is $\rho = \sqrt{8}$. The cone is $z = r$, so

$$\rho \cos \phi = \rho \sin \phi,$$

which gives $\phi = \pi/4$ away from the origin. Hence

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \rho \leq \sqrt{8}.$$

Therefore

$$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^{\sqrt{8}} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{32\pi}{3}(\sqrt{2} - 1).$$

Sphere-cone solid as a spherical box

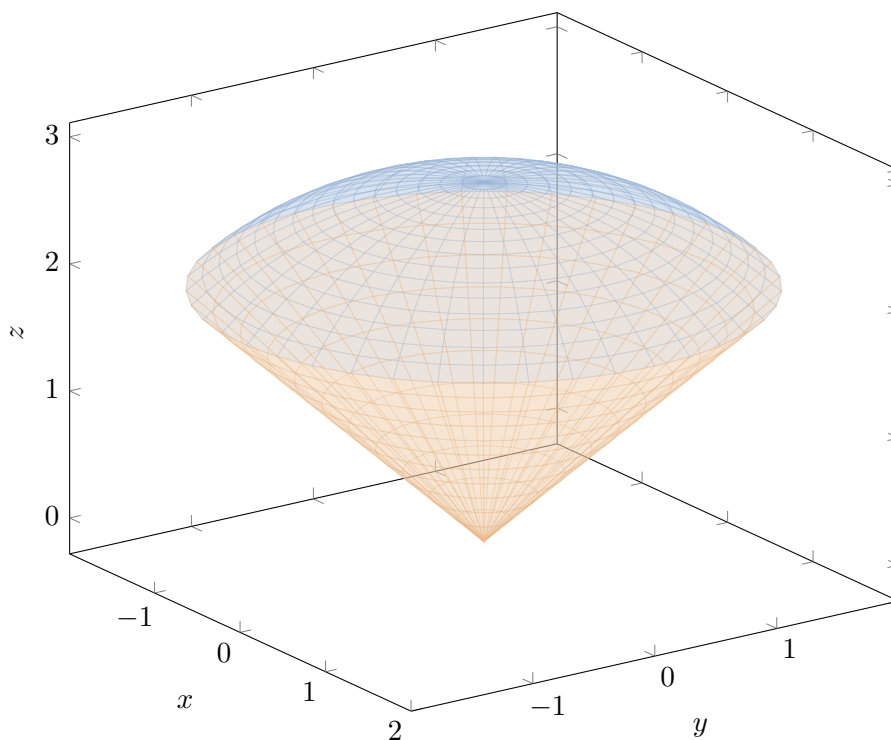


Figure 17.9: The region inside a sphere and above a cone becomes a simple rectangular box in (ρ, ϕ, θ) -space.

Example 8: A spherical shell in the first octant

Let E be the part of the spherical shell $1 \leq x^2 + y^2 + z^2 \leq 4$ in the first octant. The first octant means $x, y, z \geq 0$, so

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq \frac{\pi}{2}.$$

The shell condition gives $1 \leq \rho \leq 2$. Hence

$$V = \int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \left(\frac{\pi}{2}\right) (1) \left(\frac{2^3 - 1^3}{3}\right) = \frac{7\pi}{6}.$$

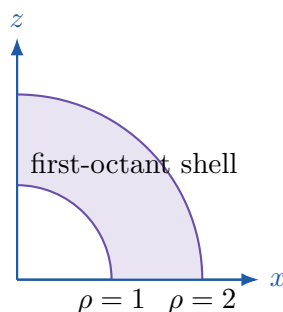


Figure 17.10: A cross-section of the first-octant spherical shell. The full three-dimensional region has $0 \leq \theta \leq \pi/2$ and $0 \leq \phi \leq \pi/2$.

17.9 Choosing among rectangular, cylindrical, and spherical coordinates

Coordinate choice is part of problem solving. A good choice makes the region and integrand simple. A poor choice may make a straightforward problem unnecessarily complicated.

Geometry or expression	Recommended coordinates	Reason
Box with constant x, y, z bounds	rectangular	boundaries already align with coordinate planes
Cylinder $x^2 + y^2 = R^2$	cylindrical	becomes $r = R$
Cone $z = c\sqrt{x^2 + y^2}$	cylindrical or spherical	becomes $z = cr$ or $\phi = \text{constant}$
Sphere $x^2 + y^2 + z^2 = R^2$	spherical	becomes $\rho = R$
Radial integrand $f(x^2 + y^2)$	cylindrical	becomes $f(r^2)$
Radial integrand $f(x^2 + y^2 + z^2)$	spherical	becomes $f(\rho^2)$
First octant of a ball	spherical	constant angle bounds
Solid of revolution around the z -axis	cylindrical	natural r and z description

A practical workflow

1. Sketch or identify the boundary surfaces.
2. Translate the boundary equations into candidate coordinate systems.
3. Choose the system where most bounds become constants or simple functions.

4. Rewrite the integrand.
5. Include the correct volume element.

17.10 Numerical integration in cylindrical and spherical coordinates

The coordinate formulas are useful not only for exact integration but also for computation. A numerical method works best when the region has simple bounds. For the sphere-cone solid from Example 7,

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{4}, \quad 0 \leq \rho \leq \sqrt{8}.$$

A grid-based spherical-coordinate sum approximates the volume by

$$\sum \rho_{ijk}^2 \sin \phi_j \Delta \rho \Delta \phi \Delta \theta.$$

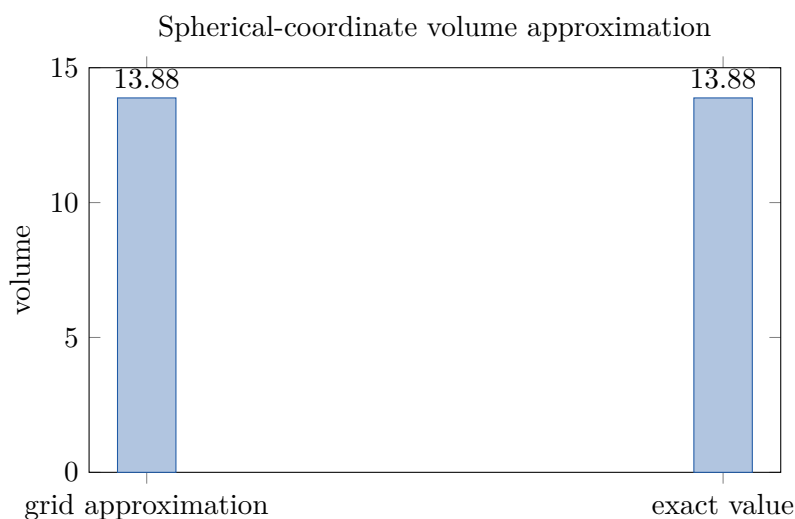


Figure 17.11: For regions with spherical bounds, a grid in (ρ, ϕ, θ) directly matches the geometry.

17.11 Coordinate transformations and probability

Triple integrals are also probability integrals. If $f(x, y, z)$ is a joint density, then

$$P((X, Y, Z) \in E) = \iiint_E f(x, y, z) dV.$$

When the density is radially symmetric, spherical coordinates are natural. For example, the three-dimensional standard normal density is

$$f(x, y, z) = \frac{1}{(2\pi)^{3/2}} e^{-(x^2+y^2+z^2)/2}.$$

Since $x^2 + y^2 + z^2 = \rho^2$, spherical coordinates give

$$f = \frac{1}{(2\pi)^{3/2}} e^{-\rho^2/2}.$$

The probability mass in a ball of radius R is

$$P(\rho \leq R) = \int_0^{2\pi} \int_0^\pi \int_0^R \frac{1}{(2\pi)^{3/2}} e^{-\rho^2/2} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

After integrating out the angles,

$$P(\rho \leq R) = \sqrt{\frac{2}{\pi}} \int_0^R \rho^2 e^{-\rho^2/2} \, d\rho.$$

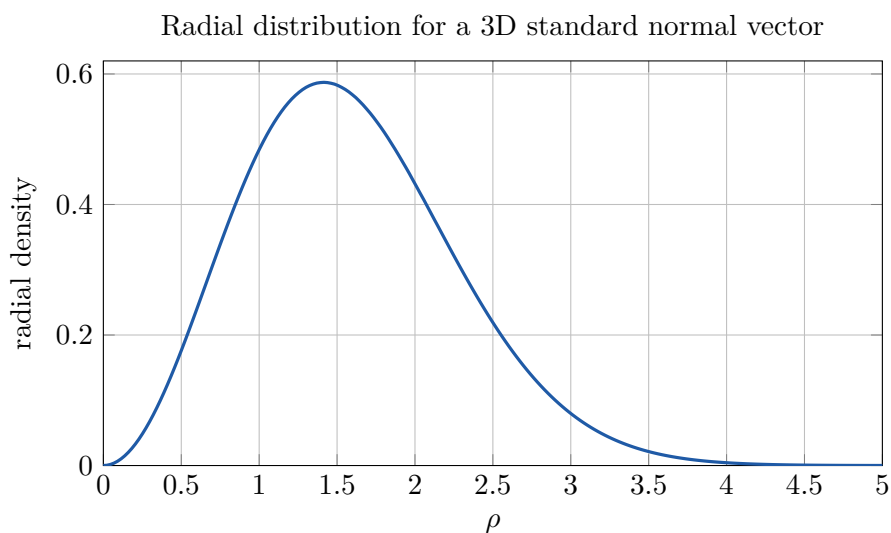


Figure 17.12: A rotationally symmetric three-dimensional density becomes a one-dimensional radial density after integrating out angular variables.

17.12 Sampling uniformly from a ball

Coordinate transformations matter in simulation. To sample uniformly from the unit ball in $\mathbb{R}^3$, it is not correct to choose ρ uniformly from $[0, 1]$. The volume element contains ρ^2 , so more volume lies near the boundary.

A correct method is

$$\theta = 2\pi U_1, \quad \cos \phi = 2U_2 - 1, \quad \rho = U_3^{1/3},$$

where U_1, U_2, U_3 are independent uniform random variables on $[0, 1]$. The power $1/3$ appears because the volume of a ball of radius r is proportional to r^3 .

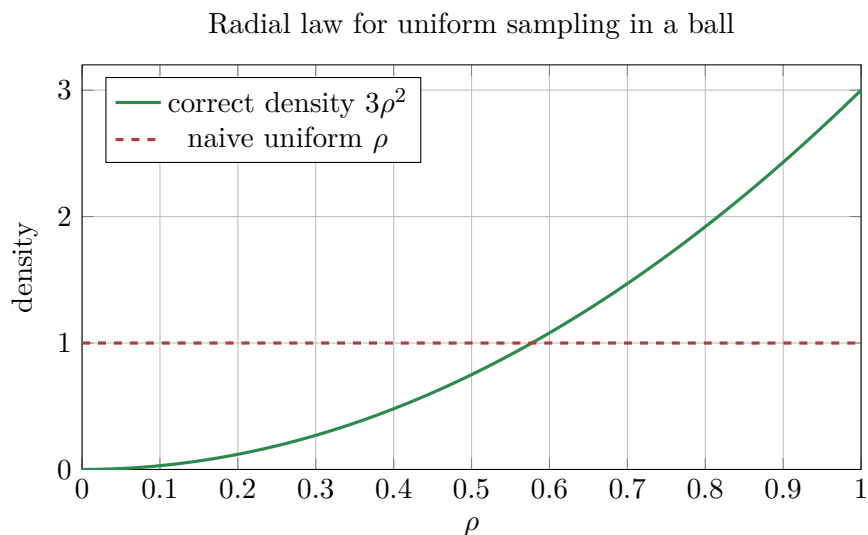


Figure 17.13: Uniform volume sampling has radial density $3\rho^2$, not a flat radial density.

17.13 Common mistakes

Mistake 1: Forgetting the Jacobian factor

In cylindrical coordinates, the volume element is $r \, dz \, dr \, d\theta$, not just $dz \, dr \, d\theta$. In spherical coordinates, the volume element is $\rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$, not just $d\rho \, d\phi \, d\theta$.

Mistake 2: Confusing r and ρ

In cylindrical coordinates, $r = \sqrt{x^2 + y^2}$. In spherical coordinates, $\rho = \sqrt{x^2 + y^2 + z^2}$. Thus r is distance to the z -axis, while ρ is distance to the origin.

Mistake 3: Confusing θ and ϕ

The angle θ is measured in the xy -plane. The angle ϕ is measured down from the positive z -axis.

Mistake 4: Writing bounds before understanding the geometry

Do not begin by guessing bounds. First translate the boundary surfaces. Then decide which coordinate system makes them simple.

17.14 Chapter summary

Cylindrical coordinates are defined by

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

with volume element

$$dV = r \, dz \, dr \, d\theta.$$

Spherical coordinates are defined by

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

with volume element

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta.$$

The main problem-solving skill is choosing coordinates that match the region and integrand. Cylindrical coordinates simplify cylinders, cones, and axial symmetry. Spherical coordinates simplify spheres, balls, shells, cones, and radial functions.

17.15 Practice problems

1. Convert the cylindrical point $(4, \pi/6, 2)$ to rectangular coordinates.
2. Convert the rectangular point $(1, 1, 3)$ to cylindrical coordinates.
3. Describe the surface $r = 5$ in rectangular coordinates.
4. Describe the surface $z = 2r$ in rectangular coordinates.
5. Find the volume inside $x^2 + y^2 = 9$ between $z = 0$ and $z = 4$.
6. Find the mass of the solid in Problem 5 if the density is $\delta(x, y, z) = x^2 + y^2$.
7. Set up a cylindrical-coordinate integral for the volume under $z = 4 - x^2 - y^2$ and above the xy -plane.
8. Evaluate the integral from Problem 7.
9. Convert the spherical point $(\rho, \theta, \phi) = (2, \pi/3, \pi/4)$ to rectangular coordinates.
10. Describe the surface $\rho = 3$ in rectangular coordinates.
11. Describe the surface $\phi = \pi/3$ in rectangular coordinates.
12. Find the volume of the ball $x^2 + y^2 + z^2 \leq 27$.
13. Find the volume of the upper hemisphere $x^2 + y^2 + z^2 \leq 9, z \geq 0$.
14. Set up a spherical-coordinate integral for the volume inside $x^2 + y^2 + z^2 = 16$ and above the cone $z = \sqrt{3(x^2 + y^2)}$.
15. Evaluate the integral in Problem 14.
16. Let E be the first-octant part of the ball $x^2 + y^2 + z^2 \leq 8$. Set up and evaluate a spherical-coordinate integral for its volume.
17. Explain why choosing ρ uniformly from $[0, 1]$ does not produce uniform points in the unit ball.
18. For the density $f(x, y, z) = Ce^{-(x^2+y^2+z^2)}$, write the normalization condition for C using spherical coordinates.

17.16 Python lab and interactive page

The lab for this chapter asks you to plot cylindrical and spherical coordinate surfaces, compare descriptions of the same solid, approximate triple integrals with coordinate grids, simulate points uniformly in a ball, and compare radial distributions.

- Python lab: `../labs/chapter-17-cylindrical-and-spherical-coordinates.ipynb`
- Interactive page: `../htmls/chapter-17-cylindrical-and-spherical-coordinates.html`